Communication Networks - NUIST

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Communication Networks - NUIST (A35618)

Short Title:	Communication Networks (NUIST)		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Discipline Specific Technologies		
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Credits	5	**Level:**	Postgraduate

Description of Module / Aims

This module concentrates on the Design, Management and Security aspects of Computer Networks. It provides a clear and effective indicative content that allows students to engage in the development of small/ medium scale LANs for residential or corporate environments.

Programmes

MEng in Electronic Engineering (`WD_EELEN_R`)	1 / 1 / E
MEng in Electronic Information Systems (`WD_EELEN_RI`)	2 / 3 / M
PG Dip in Engineering in Electronic Engineering (`WD_EELEN_G`)	1 / 1 / E

Indicative Content

- Addressing Subnetting, DHCP, NAT, IPv6
- Routing Routing Protocols and Algorithms, Routing Tables
- TCP/IP versions and performance analysis
- Security Protocols IPSEC, TLS, PGP, Firewalls, VPN
- Network Management
- Structure equipment and cabling (Gigabit Ethernet, Wireless Lans IEEE802.11n/g, Bluetooth)
- Software-defined networks
- Congestion Control & QoS
- Project-oriented and top-down network design
- Selected topics on networking at the edge (Internet of Things, Industrial Networks, Body Access Networks)

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Survey the salient features of a data communication network, including equipment, tools, standards and protocols.
2. Specify and design a small to medium size LAN.
3. Design he logic and physical LAN set up in a residential or co-operative environment.
4. Assess performance, security and scalability of LAN.
5. Optimise the LAN management and configuration.
6. Validate the designed LAN through structured professional documentation and oral presentation.

Learning and Teaching Methods

- Lectures
- Project Based Learning
- Design Assignments

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5
Continuous Assessment	50%	
Project	30%	2,5,6
Assignment	20%	1,2,3,4,5,6

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

40% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	6	
Independent Learning	105	

Essential Material

- Forouzan, B. _Data Communications and Networking_. 3rd Ed.. .: McGraw Hill, 2004.
- Forouzan, B. _tcp/ip Protocol Suite_. 2nd Ed.. .: McGraw Hill, 2003.
- Leon-Garcia, A. and I. Widjaja. _Communication Networks_. 2nd Ed.. .: McGraw Hill, 2004.
- Stallings, W. _Data and Computer Communications_. 7th Ed.. .: Prentice Hall, 2003.
- Tomasi , W. _Introduction to Data Communications_. .: Prentice Hall, 2005.